The set of infinite binary strings is not enumerable

The Open Logic Project contributors

openlogic:sets-functions-relations/size-of-sets:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author's own theorem environment at source lines 1170-1173 (the \ollabel on line 1171; the author puts the environment start on its own line); the proof environment begins two lines below the statement and is its own proof.

Editorial note. Neither the statement nor the proof contains an \olref, an \Olref or a \cref, so no source pointer and no bibliography entry are needed; every symbol used, including the binary-sequences notation and the array environment, is declared in the retained preamble.

Editorial note. The proof turns on a displayed infinite array; the array is part of the retained source range and is reproduced as drawn, so no figure file is needed.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained theorem prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's size-of-sets unit, counted once: the infinite binary sequences are not enumerable.

Allowed standard prerequisites (editorial).

- Enumerable set. A set A is enumerable when there is a surjection from the positive integers onto A, or equivalently when its members can be listed.. The proof starts from an arbitrary listing of a subset of the binary sequences and derives a sequence missing from the list (source lines 1176-1180).
- Infinite binary strings. The set of infinite sequences of 0s and 1s, written as the set of functions from the natural numbers to the two symbols.. The array of digits and the constructed string d are both infinite binary strings (source lines 1176-1200).

Theorem 0.1. Bin^{ω} *is !!nonenumerable.*

Proof. Consider any enumeration of a subset of Bin^ω. So we have some list s_0, s_1, s_2, … where every s_n is an infinite string of 0's and 1's. Let $s_n(m)$ be the nth digit of the mth string in this list. So we can now think of our list as an array, where $s_n(m)$ is placed at the nth row and mth column:

	0	1	2	3	…
0	$\mathbf{s_0(0)}$	$s_0(1)$	$s_0(2)$	$s_0(3)$	…
1	$s_1(0)$	$\mathbf{s_1(1)}$	$s_1(2)$	$s_1(3)$	…
2	$s_2(0)$	$s_2(1)$	$\mathbf{s_2(2)}$	$s_2(3)$	…
3	$s_3(0)$	$s_3(1)$	$s_3(2)$	$\mathbf{s_3(3)}$	…
⋮	⋮	⋮	⋮	⋮	⋱

We will now construct an infinite string, d, of 0's and 1's which is not on this list. We will do this by specifying each of its entries, i.e., we specify $d(n)$ for all $n \in \mathbb{N}$. Intuitively, we do this by reading down the diagonal of the array above (hence the name "diagonal method") and then changing every 1 to a 0 and every 1 to a 0. More abstractly, we define $d(n)$ to be 0 or 1 according to whether the n-th !!element of the diagonal, $s_n(n)$, is 1 or 0, that is:

$$d(n) = \begin{cases} 1 & \text{if } s_n(n) = 0 \\ 0 & \text{if } s_n(n) = 1 \end{cases}$$

Clearly $d \in \text{Bin}^\omega$, since it is an infinite string of 0's and 1's. But we have constructed d so that $d(n) \neq s_n(n)$ for any $n \in \mathbb{N}$. That is, d differs from s_n in its nth entry. So $d \neq s_n$ for any $n \in \mathbb{N}$. So d cannot be on the list s_0, s_1, s_2, …

We have shown, given an arbitrary enumeration of some subset of Bin^ω, that it will omit some !!element of Bin^ω. So there is no enumeration of the set Bin^ω, i.e., Bin^ω is !!nonenumerable. □